MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS
FALL SEMESTER 2025
TEST 1

Last Name:

First Name:

UID:

Instructons:

(1) The question paper is in the **last page**. You may tear off the last page, but **return the questions paper** at the end of the test.
(2) Write answers in the attached A4 papers. Writings in draft papers will NOT be graded.
(3) Show your steps or reasonings clearly to receive full credits unless otherwise specified in the question.

Write your work in this page

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Test Paper (Fall 2025)
Writings on this page will not be graded

Name:
UID:

(1) (4 points) Define a Cauchy sequence (use $\epsilon - N$-definition).

(2) (5 points) Let the set

$$A = \left\{ 3 \pm \frac{n}{3n+1} : n \in \mathbb{N} \right\}.$$

Find $\sup(A)$ and $\inf(A)$. Briefly explain your answers.

(3) (5 points) Find the limit

$$\lim \frac{n!}{n^n}$$

if the limit exists and justify your answer. If the limit does not exists, explain why.

(4) (6 points) Let (x_n) be a sequence. Prove that if the series $\sum x_n$ is absolutely convergent, then $\sum x_n$ is convergent. (You can use comparison test without proving it.)

(5) (4 points) Define $\mathbb{Q}(i)$ to be the set whose elements are of the form $a + bi$ for $a, b \in \mathbb{Q}$. Define additions and multiplication as follows:

(a) For $a_1, b_1, a_2, b_2 \in \mathbb{Q}$, define $(a_1 + b_1 i) + (a_2 + b_2 i) = (a_1 + a_2) + (b_1 + b_2)i$;

(b) For $a_1, b_1, a_2, b_2 \in \mathbb{Q}$, define $(a_1 + b_1 i) \cdot (a_2 + b_2 i) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i$.

The above addition and multiplication define $\mathbb{Q}(i)$ to be a field. (You do not have to check it is a field.) Explain briefly whether $\mathbb{Q}(i)$ is ordered. (Some axioms are provided in the next page.)

(6) (6 points) Let (x_n) be a bounded sequence in $\mathbb{R}$. Prove or disprove: If (x_n) is divergent, then there exists two convergent subsequences of (x_n) whose limits are not the same.

Axioms for ordered fields F:

(O1) For any $a, b \in F$, $a \leq b$ or $b \leq a$.

(O2) For $a, b \in F$, if $a \leq b$ and $b \leq a$, then $a = b$.

(O3) For $a, b, c \in F$, if $a \leq b$ and $b \leq c$, then $a \leq c$.

(O4) For $a, b, c \in F$, if $a \leq b$, then $a + c \leq b + c$.

(O5) For $a, b, c \in F$, if $a \leq b$ and $c \geq 0$, then $ac \leq bc$.